# MPD (Modèle Physique de Données) Projet Correspondances n°2

Railway_line

**Railway_line**

**line_id** BIGSERIAL
name VARCHAR(120) NOT NULL
code VARCHAR(20) NOT NULL

**PK** (line_id)
**UQ** (code)

Line_station

**Line_station**

***line_id*** BIGINT NOT NULL
***station_id*** BIGINT NOT NULL
stop_order INT NOT NULL

**PK** (line_id, station_id)
**FK** line_id → Railway_line ON DELETE **CASCADE**
**FK** station_id → Station ON DELETE **RESTRICT**
**UQ** (line_id. stop_order)
**CK** (stop_order > 0)

Station

**Station**

**station_id** BIGSERIAL
name VARCHAR(120) NOT NULL
city VARCHAR(100) NOT NULL
code VARCHAR(10) NOT NULL

**PK** (station_id)
**UQ** (code)

Train_stop

**Train_stop**

**train_stop_id** BIGSERIAL
*train_id* BIGINT NOT NULL
*station_id* BIGINT NOT NULL
stop_order INT NOT NULL
arrival_time TIME NULL
departure_time TIME NULL

**PK** (train_stop_id)
**FK** train_id → Train ON DELETE **CASCADE**
**FK** station_id → Station ON DELETE **RESTRICT**
**UQ** (train_id, stop_order)
**UQ** (train_id, station_id)
**CK** (stop_order > 0)
**CK** (arrival_time IS NOT NULL OR departure_time IS NOT NULL)

Distance_band

**Distance_band**

**distance_band_id** BIGSERIAL
min_distance_km NUMERIC(8,2) NOT NULL
max_distance_km NUMERIC(8,2) NOT NULL

**PK** (distance_band_id)
**CK** (min_distance_km >= 0)
**CK** (max_distance_km > min_distance_km)

Service_calendar

**Service_calendar**

**calendar_id** BIGSERIAL
name VARCHAR(60) NOT NULL
monday BOOLEAN NOT NULL DEFAULT FALSE
tuesday BOOLEAN NOT NULL DEFAULT FALSE
wednesday BOOLEAN NOT NULL DEFAULT FALSE
thursday BOOLEAN NOT NULL DEFAULT FALSE
friday BOOLEAN NOT NULL DEFAULT FALSE
saturday BOOLEAN NOT NULL DEFAULT FALSE
sunday BOOLEAN NOT NULL DEFAULT FALSE

**PK** (calendar_id)

Calendar_exception

**Calendar_exception**

**exception_id** BIGSERIAL
*calendar_id* BIGINT NOT NULL
service_date DATE NOT NULL
is_running BOOLEAN NOT NULL

**PK** (exception_id)
**FK** calendar_id → Service_calendar ON DELETE **CASCADE**
**UQ** (calendar_id. service_date)

Train

**Train**

**train_id** BIGSERIAL
*line_id* BIGINT NOT NULL
*calendar_id* BIGINT NOT NULL
train_number VARCHAR(20) NOT NULL

**PK** (train_id)
**FK** line_id → Railway_line ON DELETE **RESTRICT**
**FK** calendar_id → Service_calendar ON DELETE **RESTRICT**
**UQ** (train_number)

Train_circulation

**Train_circulation**

**circulation_id** BIGSERIAL
*train_id* BIGINT NOT NULL
service_date DATE NOT NULL

**PK** (circulation_id)
**FK** train_id → Train ON DELETE **RESTRICT**
**UQ** (train_id, service_date)

Reservation

**Reservation**

**reservation_id** BIGSERIAL
*circulation_id* BIGINT NOT NULL
*seat_id* BIGINT NOT NULL
*traveler_id* BIGINT NOT NULL
*departure_station_id* BIGINT NOT NULL
*arrival_station_id* BIGINT NOT NULL
departure_order INT NOT NULL
*fare_id* BIGINT NOT NULL

**PK** (reservation_id)
**FK** circulation_id → Train_circulation ON DELETE **RESTRICT**
**FK** seat_id → Seat ON DELETE **RESTRICT**
**FK** traveler_id → Traveler ON DELETE **RESTRICT**
**FK** departure_station_id → Station ON DELETE **RESTRICT**
**FK** arrival_station_id → Station ON DELETE **RESTRICT**
**FK** fare_id → Fare ON DELETE **RESTRICT**
**CK** (departure_station_id <> arrival_station_id)
**CK** (departure_order > 0)

Train_car

**Train_car**

**car_id** BIGSERIAL
*train_id* BIGINT NOT NULL
car_number INT NOT NULL

**PK** (car_id)
**FK** train_id → Train ON DELETE **CASCADE**
**UQ** (train_id. car_number)
**CK** (car_number > 0)

Seat

**Seat**

**seat_id** BIGSERIAL
*car_id* BIGINT NOT NULL
*class_id* BIGINT NOT NULL
seat_number VARCHAR(10) NOT NULL

**PK** (seat_id)
**FK** car_id → Train_car ON DELETE **CASCADE**
**FK** class_id → Seat_class ON DELETE **RESTRICT**
**UQ** (car_id. seat_number)

Seat_class

**Seat_class**

**class_id** BIGSERIAL
name VARCHAR(40) NOT NULL

**PK** (class_id)
**UQ** (name)

Traveler

**Traveler**

**traveler_id** BIGSERIAL
*profile_id* BIGINT NOT NULL
first_name VARCHAR(80) NOT NULL
last_name VARCHAR(80) NOT NULL
email VARCHAR(160) NULL

**PK** (traveler_id)
**FK** profile_id → Traveler_profile ON DELETE **RESTRICT**
**UQ** (email)

Traveler_profile

**Traveler_profile**

**profile_id** BIGSERIAL
name VARCHAR(40) NOT NULL

**PK** (profile_id)
**UQ** (name)

Fare

**Fare**

**fare_id** BIGSERIAL
*distance_band_id* BIGINT NOT NULL
*class_id* BIGINT NOT NULL
*period_id* BIGINT NOT NULL
*profile_id* BIGINT NOT NULL
price NUMERIC(10,2) NOT NULL
valid_from DATE NOT NULL

**PK** (fare_id)
**FK** distance_band_id → Distance_band ON DELETE **RESTRICT**
**FK** class_id → Seat_class ON DELETE **RESTRICT**
**FK** period_id → Fare_period ON DELETE **RESTRICT**
**FK** profile_id → Traveler_profile ON DELETE **RESTRICT**
**UQ** (distance_band_id, class_id, period_id,
profile_id, valid_from)
**CK** (price >= 0)

Fare_period

**Fare_period**

**period_id** BIGSERIAL
name VARCHAR(60) NOT NULL
start_time TIME NULL
end_time TIME NULL

**PK** (period_id)
**UQ** (name)

**Mes Contraintes**

| | |
|---|---|
| **PK** | Ma clé primaire : elle identifie chaque ligne et n'est jamais vide. |
| **FK** | Ma clé étrangère : la valeur doit déjà exister dans la table liée. |
| **UQ** | Je n'autorise aucun doublon. |
| **CK** | Une condition que chaque ligne doit respecter. |
| ON DELETE **CASCADE** | Si je supprime le parent, ses enfants partent avec. |
| ON DELETE **RESTRICT** | Je ne peux pas supprimer le parent tant qu'il est utilisé. |
| NOT NULL | Valeur obligatoire. |
| NULL | Je peux laisser vide. |